

Value-at-Risk (VaR) Forecasting Task

1 Background

Value-at-Risk (VaR) is a standard risk measure used in finance to quantify the potential loss of an asset or portfolio over a given horizon at a specified confidence level. Accurate VaR forecasts are essential for capital allocation, regulatory compliance, and internal risk management.

In this task, you will forecast the daily VaR of SPY log-returns using both classical statistical methods and neural network approaches. All models must be evaluated using a **rolling window** framework.

Please send email to **MiniProjReceiver@163.com** with the report, code and **resume**.

2 Dataset Description

The dataset contains daily observations for SPY with three variables:

- **log_ret**: logarithmic returns
- **rv5**: realized volatility
- **bv**: bipower variation

The data are daily. You are free to choose the training/validation/test split, but the rolling window design must be strictly adhered to.

3 Objective

Forecast the one-day-ahead VaR of SPY log-returns at three confidence levels:

$$\alpha \in \{1\%, 5\%, 10\%\}$$

where VaR_α satisfies $P(r_{t+1} \leq \text{VaR}_\alpha) = \alpha$.

4 Methodological Requirements

You must implement **at least one non-neural network** method and **at least one neural network** method. You are not limited to the examples listed below.

4.1 Non-Neural Approaches (examples)

- non-parametric methods (e.g. historical simulation)
- GARCH-type models (e.g., GARCH(1,1))
- Quantile regression
- other methods

4.2 Neural Network Approaches (examples)

- Multilayer perceptron (MLP)
- LSTM or GRU networks
- Quantile regression neural network (QRNN)
- Transformer-based models

5 Submission Requirements

Submit a single PDF report and the corresponding code (Python scripts) and **your CV**. The report should include:

- Data exploration
- Detailed description of each model and motivation for choosing it
- Rolling window settings and computational details
- Tables and figures of VaR forecasts, failure rates, test results with statistical hypothesis testing
- Comparative discussion: which model performs best at each tail level? How do neural methods compare to classical ones?
- Conclusion and potential improvements.
- **Please show your capabilities and unique strengths, like math ability, programming skill, or domain knowledge in the report.**

Software and Libraries (Suggested)

Python 3.8+, pandas, numpy, matplotlib, statsmodels, arch (for GARCH), scikit-learn, PyTorch.